Problem Solving

How to Write a Problem Statement

- A Problem Statement is a clear description of an issue, challenge, or gap that identifies the problem and defines the need for an effective solution and action.

What is a Problem Statement?

- A Problem Statement is a concise description of an issue that needs to be solved or improved in a project

What Makes a Good Problem Statement?

- A good Problem Statement clearly explains the issue and why it needs attention. It should define the challenge in a simple, specific, and objective way while focusing on the root cause and its impact.
- ***An effective Problem Statement often answers the five Ws to provide clarity:***

a) Who is affected by the problem?
b) What is the actual issue or gap?
c) Where does the problem occur?
d) When does the issue happen or require resolution?
e) Why is solving the problem important?

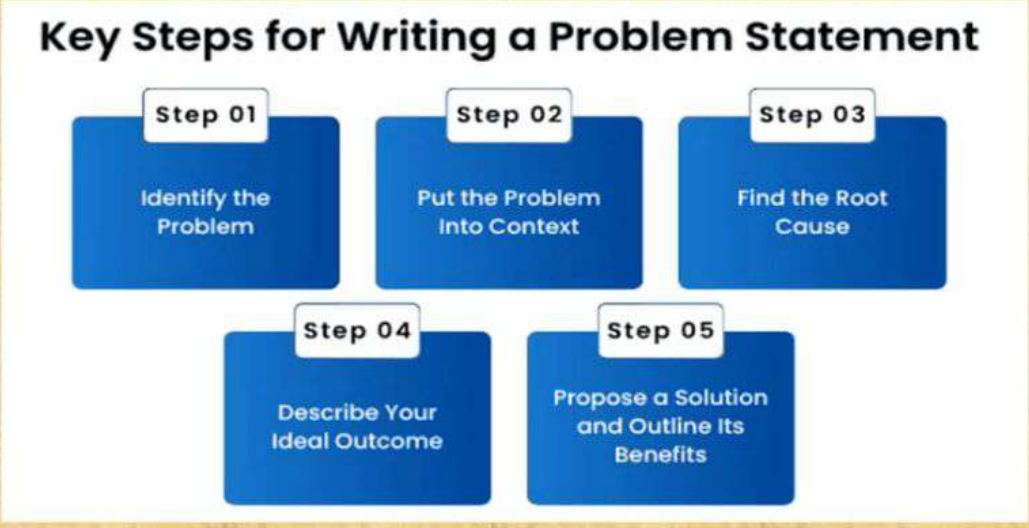
Why is a Problem Statement Important?

Problem Statements are important for the following reasons:

1. **Clarify Focus:** It provides a deep understanding of the issue, preventing teams from tackling symptoms instead of root causes.
2. **Lead Planning:** It serves as a blueprint for developing project goals and deliverables.
3. **Enhance Communication:** Stakeholders, team members, and sponsors get aligned on what problem is being solved and why.
4. **Stimulate Innovation:** With a clearly defined problem, creative and practical solutions become easier to brainstorm.
5. **Improve Resource Use:** It helps ensure that time, budget, and energy are used on the right things.
6. **Prevents Recurring Issues:** By identifying the root problem early, it reduces the chances of the same issue happening again in the future.
7. **Improves Outcome Visibility:** It outlines the problem and the expected outcome, which sets clear expectations for everyone involved.

Problem Solving

How to Write a Problem Statement?



Problem Solving

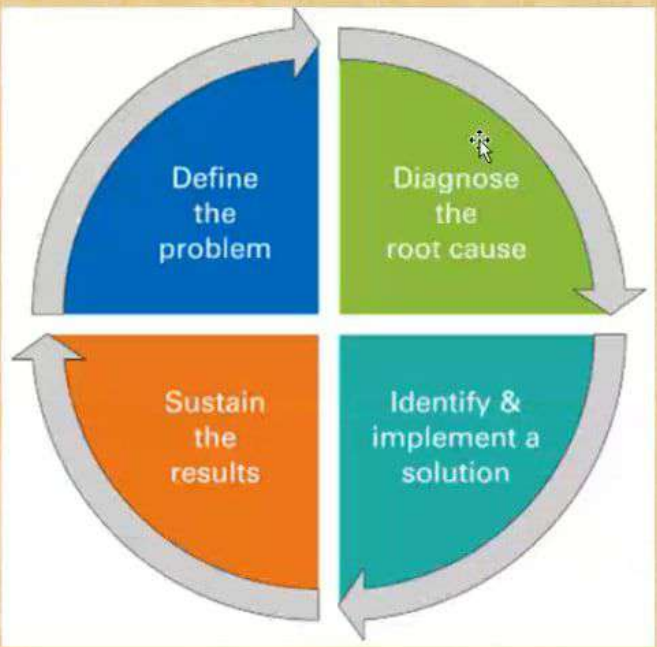
Tips for Writing a Strong Problem Statement



Problem Solving

WHAT IS PROBLEM SOLVING?

Solving a problem depends on correctly identifying its cause so the best solution can be selected and implemented for sustained results.



Problem Solving

the high level problem-solving process is:

Step 1 – Define the problem

When defining the problem—that is, stating the problem specifically—it's important to differentiate fact from opinion and discover underlying causes so the focus is on the problem, not just its symptoms. To do this, it's important to:

- Involve all interested workgroups affected by the problem and ask subject matter experts to better understand the involved processes or issues.
- Collect and analyze supporting data (qualitative and quantitative).
- Go to the gemba review and document how processes currently work (that is, who does what, with what information, using what tools, communicating with what organizations and individuals, in what timeframe, using what format).

Problem Solving

- *Answer these descriptive questions:*
- What (what happened) – What is the problem? What product is defective?
- Where (point of detection) – Where was the problem detected?
- When (date/time of detection) – When was the problem detected? When does the problem occur?
- How (many/much) – How many products are affected? How much of production (in percentage) is defective?
- Who (detection only, not cause) – Who detected the problem?
- Why (high level) – Why is this a problem?

- ***Step 2 – Diagnose the root cause***

Root cause analysis (RCA) is a collective term that describes a wide range of approaches, tools, and techniques used to uncover causes of problems and direct corrective action. RCA also can be one part of a problem-solving technique, as with the A3 report.

Some guidance questions for determining a root cause include:

- Why? Why? Why? Why? Why? (also known as the 5 whys technique)
- What other causes were studied and eliminated?
- How did you verify this cause?

- ***Step 3 – Identify and implement a solution***

Generate potential solutions

Postpone selecting one solution until several alternatives have been proposed—for example short- and long-term solutions. Considering multiple alternatives can enhance the value of your ideal solution significantly. All too often, groups start planning to carry out their first idea. Play with additional ideas, and you probably will improve on your first one. Now is the time to think boldly and creatively —for example, by combining two merely OK solutions into one hybrid great solution, or asking customers and suppliers to help develop solutions to problems that affect them.

Problem Solving

Consider the following:

- What potential solutions have you evaluated?
- What methods did you use to encourage creative solutions?
- Are there any constraints preventing you from choosing a solution?
- How does this solution address the root cause?
- What are the possible problems with this solution? What are your contingency plans?
- How will you communicate your planned solution?
- How will you monitor your plan?
- How will you know whether the solution is successful?

Problem Solving

Plan the change and implement it

When a solution is selected, implementation will be easier with a carefully developed plan, including details on what, who, and when.

A carefully developed plan also should determine how the solution will be measured to determine the results of the change and to monitor how well the plan was implemented. Be sure to include contingency planning in case something falters using case analysis tools to develop these contingencies. These include:

- Identifying potential obstacles or barriers (why and/or what) and deciding how to avoid or overcome them.
- Planning and documenting how you will react in the worst-case scenario.
- Considering the human elements that may affect your success, such as who will feel threatened and whether the new procedures will be difficult to remember.

Problem Solving

Planning

- Planning is a process of organizing pieces of knowledge into a consistent sequence of actions that will accomplish a goal.
- The task of a planner is to find a sequence of actions that allow a problem solver, such as a control system, to accomplish some specific task.
- Traditional planning is very much knowledge-intensive, since plan creation requires the organization of pieces of knowledge and partial plans into a solution procedure.
- Planning plays a role in expert systems in reasoning about events occurring over time.
- Planning has many applications in manufacturing, such as process control. It is also important for designing a robot controller.

Planning Examples:

- Identify the actions and the task or goal n
- Sequence of actions to go get block (a) from room (b)

1. Put down whatever is now held
2. Go to room b
3. Pick up block a
4. Leave room b
5. Return to original location

Problem Solving

- The steps of a traditional robot are composed of the robot's *atomic actions*
- *Atomic actions* can be at the micro-level or higher level
- Plans are created by searching through a space of possible actions until the sequence necessary to accomplish the task is discovered.
- This space represents states of the world that are changed by applying each of the actions.
- The search terminates when the goal state (the appropriate description of the world) is produced.
- Thus, many of the issues of heuristic search, including finding A* algorithms, are also appropriate in planning.

Problem Solving

- Traditional planning relies on search techniques.
- Planning may be seen as a state space search
- The techniques of graph search may be applied to find a path from the start state to the goal state. The operations on this path constitute a plan.

- Planning Example
- *Robot's world to a set of blocks on a tabletop*

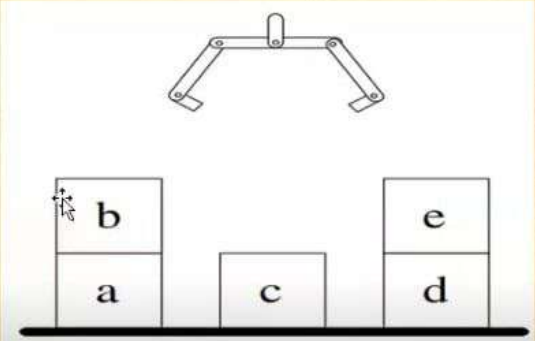


Figure 8.18 The blocks world.

Problem Solving

- We have five blocks, labeled a, b, c, d, e, sitting on the top of a table.
- The blocks are all cubes of the same size, and stacks of blocks, as in the figure, have blocks directly on top of each other.
- The robot arm has a gripper that can grasp any clear block (one with no block on top of it) and move it to any location on the tabletop or place it on top of any other block (with no block on top of it).

Problem Solving

- *The robot arm can perform the following tasks (U, V, W, X, Y, and Z are variables)*

Task	Meaning
goto(X,Y,Z)	Go to location described by coordinates X, Y, and Z. This location might be implicit in the command pickup (W) where block W has location X, Y, Z.
pickup(W)	Pick up block W from its current location. It is assumed that block W is clear on top, the gripper is empty at the time, and the computer knows the location of block W.
putdown(W)	Place block W down at some location on the table and record the new location for W. W must be held by the gripper at the time.
stack(U,V)	stack(U,V) Place block U on top of block V. The gripper must be holding U and the top of V must be clear of other blocks.
unstack(U,V)	Remove block U from the top of V. U must be clear of other blocks, V must have block U on top of it, and the hand must be empty before this command can be executed.

Problem Solving

- The state of the world is described by a set of predicates and predicate relationships:

1. location(W,X,Y,Z) Block W is at coordinates X, Y, Z.
2. on(X,Y) Block X is immediately on top of block Y.
3. clear(X) Block X has nothing on top of it.
4. gripping(X) The robot arm is holding block X.
5. gripping( ) The robot gripper is empty.
6. ontable(W) Block W is on the table.

Problem Solving

- *STRIPS, developed at what is now SRI International, stands for STanford Research Institute* Planning System.
- This controller was used to drive the *SHAKEY* robot of the early 1970s.
- The book presents a version of STRIPS-style planning and *triangle tables* (data structure) used to organize and store macro operations.
- Using blocks example, the four operators pickup, putdown, stack, and unstack are represented as triples of descriptions.

Problem Solving

- The first element of the triple is the set of *preconditions (P),* or conditions the world must meet for an operator to be applied.
- *The* second element of the triple is the *add list (A), or the additions to the state description that* are a result of applying the operator.
- Finally, there is the *delete list (D), or* the items that are removed from a state description to create the new state when the operator is applied.

Problem Solving

1- pickup(X)

P: gripping( ) ∧ clear(X) ∧ ontable(X)

A: gripping(X)

D: ontable(X) ∧ gripping( )

2- putdown(X)

P: gripping(X)

A: ontable(X) ∧ gripping( ) ∧ clear(X)

D: gripping(X)

3- stack(X,Y)

P: clear(Y) ∧ gripping(X)

A: on(X,Y) ∧ gripping( ) ∧ clear(X)

D: clear(Y) ∧ gripping(X)

4- unstack(X,Y)

P: clear(X) ∧ gripping( ) ∧ on(X,Y)

A: gripping(X) ∧ clear(Y)

D: gripping( ) ∧ on(X,Y)

- Using blocks example, the four operators pickup, putdown, stack, and unstack are represented as triples of descriptions.

Problem Solving

Decision-Making and Problem-Solving

- Effective decision-making and problem-solving are essential skills in both personal and professional realms. These processes not only help in navigating complex situations but also in achieving successful outcomes.

What is the Decision-making Process?

- The decision-making process is a systematic approach to choosing a course of action among multiple alternatives. It involves identifying a problem or opportunity, gathering relevant information, and evaluating potential solutions based on specific criteria such as feasibility, risk, and impact.

What is the Problem-Solving Process?

- The problem-solving process is a methodical approach used to identify, analyze, and resolve issues or challenges. It begins with clearly defining the problem, ensuring that the issue is well understood and framed accurately.

Problem Solving

What are Decision-making Skills?

Key decision-making skills include:

1. *Critical Thinking*
2. *Problem-Solving*
3. *Analytical Skills*
4. *Emotional Intelligence*
5. *Time Management*
6. *Risk Assessment*
7. *Time Management*
8. *Confidence*
9. *Communication*
10. *Negotiation and Persuasion*

Problem Solving

What are Problem-solving Skills?

Key problem-solving skills include:

1. Analytical Thinking
2. Creativity
3. Critical Thinking
4. Research Skills
5. Decision-Making
6. Communication
7. Collaboration
8. Adaptability
9. Perseverance
10. Attention to Detail

➢ *Importance of Problem-Solving Skills*

1. Effective Resolution of Challenges
2. Improved Efficiency and Productivity
3. Innovation and Creativity
4. Personal and Professional Growth
5. Improved Relationships

➢ *Importance of Decision-Making Skills*

1. Guiding Personal and Professional Life
2. Strategic Planning
3. Risk Management
4. Efficiency and Productivity
5. Adaptability and Resilience

Problem Solving

- *Steps in the Decision-Making Process include:*

1. *Identify the Problem or Decision to Be Made*
2. *Gather Relevant Information*
3. *Identify Alternatives*
4. *Weigh the Evidence*
5. *Choose Among Alternatives*
6. *Take Action*
7. *Review the Decision and Its Consequences*

Problem Solving

Problem-Solving Process

1. Define the Problem
2. Analyze the Problem
3. Generate Possible Solutions
4. Evaluate and Select Solutions
5. Implement the Solution
6. Monitor and Evaluate the Solution
7. Reflect and Learn